

explain Jaccard edge overlap or Jaccard Intersections



Jaccard Edge Overlap / Jaccard Intersection

Jaccard edge overlap measures how similar the **edges/connections of two graphs or two network layers** are.

It is based on the **Jaccard Similarity** between two sets.

1. Basic Formula

Suppose we have two edge sets:

- Graph/layer 1: E_1
- Graph/layer 2: E_2

Then:

$$J(E_1, E_2) = \frac{|E_1 \cap E_2|}{|E_1 \cup E_2|}$$

Where:

- $E_1 \cap E_2$ = edges present in **both** graphs
- $E_1 \cup E_2$ = edges present in **at least one** graph
- $|\cdot|$ = number of elements in the set

The value is between **0 and 1**.

Jaccard value	Meaning
0	No common edges
0.25	Low overlap
0.50	Moderate overlap
0.75	High overlap
1	Exactly the same edges

2. Simple Example

Consider two network layers.

Layer A:

$$E_A = \{(1, 2), (1, 3), (2, 3), (3, 4)\}$$

Layer B:

$$E_B = \{(1, 2), (1, 3), (3, 4), (4, 5)\}$$

Intersection

Common edges are:

$$E_A \cap E_B = \{(1, 2), (1, 3), (3, 4)\}$$

Therefore:

$$|E_A \cap E_B| = 3$$

Union

All unique edges are:

$$E_A \cup E_B = \{(1, 2), (1, 3), (2, 3), (3, 4), (4, 5)\}$$

Therefore:

$$|E_A \cup E_B| = 5$$

So:

$$J(E_A, E_B) = \frac{3}{5} = 0.6$$

Answer

$$\boxed{Jaccard\ Edge\ Overlap = 0.6}$$

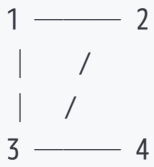
This means the two layers have **60% edge similarity** according to the Jaccard measure.

3. Why is it called "Edge Overlap"?

Imagine two people creating social networks.

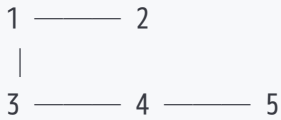
Network 1

text



Network 2

text



The edge (1,2) , (1,3) and (3,4) appear in both networks.

Those are the **overlapping edges**.

So Jaccard tells us:

Out of all unique edges appearing in either network, what fraction appear in both?

4. Jaccard Intersection vs Jaccard Similarity

These terms are sometimes used loosely, but there is an important distinction.

Jaccard Intersection

Usually refers to:

$$E_1 \cap E_2$$

This is simply the **set of common edges**.

For example:

$$E_1 \cap E_2 = \{(1,2), (1,3), (3,4)\}$$

The number of common edges is:

$$|E_1 \cap E_2| = 3$$

Jaccard Similarity

Uses the intersection **and** union:

$$J = \frac{|E_1 \cap E_2|}{|E_1 \cup E_2|}$$

So:

Intersection = common edges

Jaccard similarity = normalized common edges

5. In Multiplex Networks

This becomes particularly useful for **multiplex/multilayer networks**.

Suppose you have:

- Layer 1 → friendship network
- Layer 2 → collaboration network
- Layer 3 → communication network

Each layer has its own edges.

For example:

$$E_{friendship}$$

and

$$E_{collaboration}$$

You can calculate:

$$J(E_{friendship}, E_{collaboration})$$

to determine how much the two types of relationships overlap.

Example

Suppose:

- Friendship layer has 100 edges
- Collaboration layer has 80 edges
- 30 edges exist in both

The union is:

$$100 + 80 - 30 = 150$$

Therefore:

$$J = \frac{30}{150} = 0.20$$

So the two layers have a **20% Jaccard edge overlap**.

6. Important: Why Not Just Divide by Number of Edges?

You might think:

$$\frac{\text{common edges}}{\text{edges in Layer 1}}$$

But this would be **asymmetric**.

For example:

$$\frac{30}{100} = 0.30$$

while:

$$\frac{30}{80} = 0.375$$

Which layer should be the denominator?

Jaccard solves this by using the **union**:

$$\frac{\text{common edges}}{\text{unique edges across both layers}}$$

Therefore, it is **symmetric**:

$$J(E_1, E_2) = J(E_2, E_1)$$

7. Edge Overlap vs Jaccard Edge Overlap

Be careful because "edge overlap" can sometimes refer to different metrics.

Simple edge overlap

Could mean simply:

$$|E_1 \cap E_2|$$

Example:

There are 30 overlapping edges.

Jaccard edge overlap

Normalizes the overlap:

$$\frac{|E_1 \cap E_2|}{|E_1 \cup E_2|}$$

Example:

There are 30 common edges out of 150 unique edges, giving a Jaccard overlap of 0.20.

8. For Your Multiplex Graph Dataset

Since you're working with **multiplex .edges datasets**, Jaccard edge overlap can be used to create an $L \times L$ **layer similarity matrix**.

For example, with 4 layers:

	Layer 1	Layer 2	Layer 3	Layer 4
Layer 1	1.00	0.42	0.15	0.30
Layer 2	0.42	1.00	0.25	0.18
Layer 3	0.15	0.25	1.00	0.51
Layer 4	0.30	0.18	0.51	1.00

The diagonal is always:

$$J(E_i, E_i) = 1$$

because a layer has 100% overlap with itself.

This matrix can then be visualized as a **heatmap/network**, where higher values indicate layers with more similar edge structures.

In one sentence

Jaccard edge overlap measures the proportion of unique edges shared by two network layers: common edges divided by all unique edges appearing in either layer.



If you're using **Jaccard edge overlap for multiplex networks**, the key idea is:

$$J_{ij} = \frac{|E_i \cap E_j|}{|E_i \cup E_j|}$$

where E_i and E_j are the edge sets of two layers.

For example, if:

- Layer A = 100 edges

- Layer B = 80 edges
- Common edges = 30

then:

$$|E_A \cup E_B| = 100 + 80 - 30 = 150$$

and:

$$J_{AB} = \frac{30}{150} = 0.20$$

So **20% of the unique edges are shared between the two layers.**

Why it's useful for your dataset

If your multiplex dataset has L layers, calculate Jaccard for every pair:

$$J_{12}, J_{13}, \dots, J_{1L}, J_{23}, \dots$$

This gives an **edge-overlap matrix**:

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	Layer 1	Layer 2	Layer 3
Layer 1	1.00	0.42	0.15
Layer 2	0.42	1.00	0.31
Layer 3	0.15	0.31	1.00

- **1.0** → identical edge sets
- **0.7** → very high overlap
- **0.3** → moderate/low overlap
- **0.0** → no common edges

If you give me one of your `.edges` files, I can also explain **exactly how to calculate Jaccard edge overlap from its multiplex format**, including the meaning of each column.

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are we taking about inter layer or intra layers edges

For **Jaccard edge overlap between layers**, we are talking about **inter-layer comparison of intra-layer edges**.

The distinction

In a multiplex network:

- **Intra-layer edges** → connections **within the same layer**
 - Example: Layer 1 has edge $A - B$
 - Layer 2 has edge $A - C$
- **Inter-layer edges** → connections **between different layers**
 - Example: node A in Layer 1 connected to node A in Layer 2.

Jaccard Edge Overlap

When we calculate:

$$J(E_1, E_2) = \frac{|E_1 \cap E_2|}{|E_1 \cup E_2|}$$

we take the **intra-layer edges from Layer 1 and Layer 2** and compare them.

So:

Jaccard edge overlap measures how much the intra-layer edge sets of two different layers overlap.

For example:

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Layer 1

A — B
|
C

Layer 2

A — B
|
C — D

Layer 1:

$$E_1 = \{AB, AC\}$$

Layer 2:

$$E_2 = \{AB, AC, CD\}$$

Common intra-layer edges:

$$E_1 \cap E_2 = \{AB, AC\}$$

Union:

$$E_1 \cup E_2 = \{AB, AC, CD\}$$

Therefore:

$$J = \frac{2}{3} = 0.667$$

Important distinction

Concept	What it measures
Jaccard edge overlap	Similarity of intra-layer edges between two layers
Inter-layer edges	Connections across different layers
Jaccard intersection	Common edges between two edge sets

So if your assignment/research paper says "**Jaccard edge overlap between layers**", you generally **do not include inter-layer edges in the Jaccard calculation**. You compare the edge sets belonging to each layer.

If you show me the format of your `.edges` file, I can tell you **exactly which columns represent intra-layer and inter-layer edges in your specific dataset**.